

THEORY AND METHODS
AN EMPIRICAL COMPARISON OF MAXIMUM
LIKELIHOOD AND BAYESIAN ESTIMATION
METHODS FOR MULTIVARIATE DISEASE
MAPPING

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Summary: There are several publications comparing estimating methods for fitting a univariate spatial model. Such comparisons are rare for multivariate spatial models, which have been the focus of recent research in epidemiology. Using mortality data from the Greater Glasgow Health Board, we compare a full Bayesian approach with a frequentist approach for estimating relative risks of two diseases simultaneously. The regression parameters are estimated similarly under the two methods. Consistent with previous comparisons for univariate disease mapping, variance estimates are generally attenuated under a frequentist method. While a frequentist approach has computational limitations, a Bayesian approach, through a Markov Chain Monte Carlo (MCMC) algorithm, is less cumbersome to implement. In addition, a full Bayesian approach allows complex and flexible hierarchical modelling, and it provides more reliable estimates and predictions for many realistic epidemiological problems. The advent of the freely available WinBUGS package and the inclusion of MCMC methods in packages such as MLwiN should make Bayes methods easily accessible to most epidemiologists to

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incorporate prior information about the unknown parameters into their models.

1. Introduction

The use of multivariate disease modelling to simultaneously estimate relative risks of two or more diseases across small geographical areas is of both methodological and epidemiological importance. Such models are analytically appealing relative to performing pairwise correlations between different disease outcomes, or using some diseases as predictors in the other individual models. Joint disease models also preclude the problem of multiple testing on the same data. By pooling all the available data from different disease sources, precision and efficiency of the parameter estimates are improved. In linking specifications of the unstructured and structured spatial random effects, these models may also help in reducing identifiability problems associated with estimation of random effect parameters. From an epidemiological perspective, such models provide information on which diseases share the same risk factors, as well as identifying differences in the risk among diseases (Dabney & Wakefield, 2005, Feltbower et al., 2005, Knorr-Held & Best, 2001 and Held et al., 2005). Multiple modelling of diseases can also be used to identify disease-specific risk factors, which otherwise would have been masked by well-known common factors. Furthermore, a joint model provides estimates and inferences on the pairwise- and cross-covariances between risks of disease outcomes (Langford et al., 1999 and Leyland et al., 2000). An illustration of different ways in which diseases can co-occur either as a result of a true underlying or an artefactual mechanism can be found in Fenn, Morris and

Black (2005). An analysis of co-occurrence of diseases can be done at the subject level with a multivariate dichotomous outcome. For instance, Fenn, Morris and Black (2005) investigate pairwise co-occurrence of childhood illnesses in northern Ghana using separate bivariate probit models. However, in the current work, we concentrate on the analysis of aggregated disease count at small geographical areas, using a Poisson random effect model, where the random effects are given a multivariate normal distribution to account for possible disease co-occurrence.

There are different modelling strategies for measuring relative risks in two or more diseases. Knorr-Held and Best (2001) and Held et al. (2005) explored what is called a *shared-component model*. Under this model, the risk profile for two or more diseases is partitioned into disease-specific and shared components. Related to this, is the *latent spatial factor model* as explicated in Wang and Wall (2003), which was followed up in Liu, Wall and Hodges (2005) to handle the case of multivariate latent spatial factors. While the shared component model is based on a Poisson model, the *proportional mortality model* proposed by Dabney and Wakefield (2005) uses a binomial model for the distribution of the counts for one disease, conditional on the sum of all the two disease counts. The proportional mortality model is ideally suited when the only available data are counts of disease occurrence, but not the population at risk. Kim, Sun and Tsustakawa (2001) demonstrated the use of twofold conditional autoregressive (CAR) models for improving estimates in the analysis of bivariate disease outcomes. Alternatively, others have simply opted to model the area-specific vector of different disease log-relative risks as arising from a multivariate normal distribution (Assunção and Castro, 2004). A two-way comparison of the *shared-component* and the proportional mortality models using full Bayesian estimation methods can be found in

Dabney and Wakefield, (2005). A comparison of different statistical models for multivariate disease mapping, looking at the similarities and differences in the assumptions and inferences can be found in Manda (2006). A more general and versatile class of models than that represented in Kim, Sun and Tsutakawa (2001), Dabney and Wakefield (2005) and references therein are the multivariate CAR (MCAR) models, which can handle the case of more than two diseases (Jin, Carlin and Banerjee, 2005). These are practical and account for association between different diseases within areal units as well as the spatial association between areal units, and are based on Mardia (1988) theoretical work on multivariate Gaussian Markov Random Field (MRF). In here, we are concerned with a comparison of a full Bayesian approach with a maximum likelihood approach to fitting a particular multivariate spatial disease model as described in the Methods Section.

Research on the joint analysis of relative risks of a number of diseases has appeared only recently. We briefly describe different approaches that have so far been used for estimation. From a classical viewpoint, Langford et al. (1999) and Leyland et al. (2000) framed the multivariate spatial model as a multiple membership (MM) model (Browne, Goldstein and Rasbash, 2001) within a multilevel structure and used the iterative generalized least squares (IGLS) estimation method to fit it. The IGLS estimation method, which estimates a quasi-likelihood function, like other maximum likelihood-based methods, produces empirical Bayes (EB) estimates for the random effect parameters. The EB estimates are just the conditional means of the random effect distributions, given the overall observed data and hyperparameters, which must be estimated marginally from the data. Since uncertainty in the hyperparameters is not allowed in the modelling process, the EB methods underestimate the variability in the random effects. Thus, confidence intervals

from EB methods are narrow, leading to apparent significant results. In the case of univariate disease mapping, a comparison of empirical and full Bayesian estimation methods can be found in Heisterkamp, Doornbos and Gankema (1993) and Macnab et al. (2004) and a review of different empirical Bayes estimation methods can be found in Leyland and Davies (2005). The advantage of a full Bayesian approach relative to the marginal approach is that it accounts for the variability of all parameters in the model. The main disadvantage is the extra computational burden that may be necessary to obtain estimates and predictions. Full Bayesian methods have been used to estimate relative risk for multiple diseases simultaneously using the shared-component model (Knorr-Held and Best, 2001; Held et al., 2005) or the proportionality mortality model (Dabney and Wakefield, 2005) or simply a multivariate normal model on a set of disease-specific log-relative risks (Assunção and Castro, 2004).

Currently, the multilevel modelling software *MLwiN 2.0* (Rasbash et al., 2004) can fit univariate spatial models with both unstructured and structured spatial random terms using IGLS estimation of a quasi-likelihood function, thus providing a frequentist approach. However, extensions to multivariate spatial models are not yet available. Thus, special macros within *MLwiN* need to be written to perform the task of estimating relative risks of two or more diseases jointly (Langford et al., 1999; Leyland et al., 2000), a task not easily implementable. For Bayesian estimation, Gibbs sampling (Smith and Roberts, 1993) as implemented in the software *WinBUGS* (Spiegelhalter et al., 2004) could be coded to perform analyses for multivariate spatial models, albeit with some degree of sophistication (Feltbower et al., 2005). This paper compares a frequentist estimation to a full Bayesian estimation for fitting a bivariate spatial

disease mapping model, using, as an example, postcode mortality data from two diseases in the Greater Glasgow Health Board.

2. Methods

Suppose O_{ij} is the observed count in area i of disease j , ($i = 1, \dots, I; j = 1, 2, \dots, J$). In a standard set-up, it is assumed that O_{ij} is a Poisson random variable with mean equal to $\lambda_{ij} = E_{ij}R_{ij}$, where E_{ij} and R_{ij} are the known expected number of cases and unknown relative risk, respectively, in area i for cause j . As in random effect models, the number of cases O_{ij} are conditionally independent given the relative risks R_{ij} . Under these distributional assumptions, the maximum likelihood estimate of the relative risk R_{ij} is the usual standardized morbidity rate (SMR), $\hat{R}_{ij} = O_{ij}/E_{ij}$. It is well known that these crude risk ratios are misleading, particularly when diseases are rare or areas are small. Thus, more reliable estimates of relative risks for rare diseases or in small areas can be obtained by allowing for extra-Poisson variation (overdispersion), which is, usually, accomplished by specifying a gamma prior distribution on the relative risks R_{ij} (Clayton and Kaldor, 1987). Inclusion of extra-Poisson variation into the model accounts for unmeasured covariates, which induce spatial dependence in the O_{ij} , but it does not allow for spatial dependence between them. The latter may arise, for example, through lesser variability of rates in neighbouring densely populated urban areas as opposed to sparsely populated rural areas or through an infectious aetiology of the disease. Such explicit spatial dependence may be incorporated into the model by including a spatially structured random effect, v_{ij} , in addition to an unstructured random effect, u_{ij} . A popular model, which allows for both unstructured (heterogeneity) and structured spatial effects to be estimated independently was developed by Besag, York and Mollie (1991) (the

BYM model). The BYM model can be written as

$$\log(\lambda_{ij}) = \log(E_{ij}R_{ij}) = \log(E_{ij}) + \alpha_{0j} + \beta_j^T x_i + u_{ij} + v_{ij}$$

where α_{0j} is an intercept on the log relative risk scale for disease j in area i ; x_i is a covariate risk vector with the corresponding disease-specific coefficient parameter vector β_j . An exchangeable normal prior for the unstructured heterogeneity is easily specified as $u_{ij} \sim \text{Normal}(0, \sigma_{uj}^2)$. However, the specification for the spatial random effects is a bit more complex. A common specification is based on the conditional intrinsic Gaussian autoregressive (CAR) prior (Besag et al., 1991; Mollie, 1996). In the CAR model, the random effect v_{ij} has expected value as the average of the neighbouring random effects, hence they are not independent, a slight contrast to the multiple membership model described below. The CAR prior will, in general, produce similar estimates for v_{ij} and v_{kj} if areas i and k are spatially close. We do not fit the BYM model here. However, it suffices to point out that the multivariate version of the intrinsic CAR prior is now implemented as a standard distribution in the latest version of the *WinBUGS* software.

In this study, the specification of the spatial random effect v_{ij} is done using a multiple membership (MM) model, where the neighbourhood forms a multiple membership classification set of independent random effects $e_{ij} \sim \text{Normal}(0, \sigma_{ej}^2)$ as described in Browne, Goldstein and Rasbash (2001) (see also, Langford et al., 1999 and Leyland et al. 2000). Under this specification, the spatial random effect v_{ij} is assumed to arise as a weighted sum given by

$$v_{ij} = \sum_{k \neq i} w_{ik} e_{kj}. \quad (1)$$

The term e_{kj} can be thought of as representing the effect of area k on the other areas for disease j , weighted by w_{ik} indicating the importance of area

k to the i^{th} area. The w_{ik} could be formulated to account for the decreasing relevance of far away areas. The simplest formulation uses the adjacency model defined as $w_{ik} = 1/n_i$ if areas i and k share a common boundary and 0 otherwise, where n_i is the number of neighbours of area i .

Now, for each area i , let $u_i = (u_{i1}, u_{i2}, \dots, u_{iJ})$ and $e_i = (e_{i1}, e_{i2}, \dots, e_{iJ})$. The overall area-specific vector $\phi_i = (u_i, e_i)$ is assumed to be an independent draw from a multivariate normal distribution with mean $\mu = (0, 0, \dots, 0)$ and a $2J \times 2J$ covariance matrix Σ , with diagonal elements $(\sigma_{u1}^2, \sigma_{u2}^2, \dots, \sigma_{uJ}^2, \sigma_{e1}^2, \sigma_{e2}^2, \dots, \sigma_{eJ}^2)$ and off-diagonal elements $(\sigma_{u1u2}^2, \sigma_{u1u3}^2, \dots, \sigma_{u1eJ}^2, \sigma_{u2u1}^2, \dots, \sigma_{u2eJ}^2, \dots, \sigma_{uJeJ}^2)$ which provide easy interpretations for variances and pairwise covariances within and between areas.

Langford et al (1999) showed how to use a quasi-likelihood function to obtain the frequentist MM model estimates for both fixed effect and covariance parameters. In a Bayesian framework, the model parameters can be fitted using MCMC methods (Feltbower et al., 2005). In particular, for the Bayesian implementation, the precision matrix Σ^{-1} was given a *Wishart*(ν, Ω) prior distribution, having parameters ν , the degrees of freedom and Ω , the scale matrix. Setting the prior degrees of freedom ν equal to the order of the matrix Σ^{-1} corresponds to a minimally informative prior for the random effect variance matrix. Computations for Gibbs sampling were carried out in the software *WinBUGS* (Spiegelhalter et al., 2004), whose form for the Wishart distribution has expectation $E(\Sigma^{-1})$ equal to $\nu\Omega^{-1}$; equivalently Ω/ν is the prior guess at the covariance matrix Σ . We should also note in passing that there is great debate on choosing an appropriate specification to represent minimal prior information for the Wishart distribution. In the application, we will be looking for the relative magnitude and significance of the spatial

correlation $\sigma_{e_j e_{j'}}$ to indicate the possibility of missing spatially correlated covariates shared by two diseases.

3. Application to Scotland Cancer and Circulatory Mortality Data

3.1 Data

We compared the two estimating approaches to neoplasm and circulatory disease mortality data from the Greater Glasgow Health Board. Though the data is fairly well worn, our contribution is on the empirical comparison of a particular multivariate model for disease mapping that we present in this paper. The data consists of observed death counts from neoplasm and circulatory disease in the 143 postcode sectors of the Greater Glasgow Health Board in 1993. These data have been analysed by, among others, Langford et al. (1999), and Leyland et al. (2000). The death records were extracted from the General Registrar's Office (Scotland) and the corresponding addresses and postcodes at the time of death were linked to one of the 143 postcode sectors in existence in Greater Glasgow at the time of the 1991 Census. Population data from the 1991 Census broken down into 5-year age bands showed a total population of 922122 for the studied sectors, with mean 6448 and range 82 to 12799. In 1993, there were 3189 deaths from neoplasm and 5734 from circulatory disease in Greater Glasgow Health Board which, generally, has a poorer health experience than all other boards in Scotland (Leyland et al., 2000). The range of cancer and circulatory deaths per sector ranged from 0 to 61 and 0 to 124, respectively. Within each sector area, expected cases were estimated by fitting regression models to the age and sex subgroups. For each postcode, a measure of deprivation, the Carstairs score, was also derived. The Carstairs index of deprivation is calculated by summing the separately standardised ward-level

proportions of each of four census measures: (i) the number of unemployed males as a proportion of the total economically active male residents aged over 16; (ii) the number of persons in households with 1 or more persons per room, as a proportion of all residents in households; (iii) the number of residents in households with no car, as a proportion of all residents in households; (iv) and residents in households with an economically active head of household in social classes IV or V, as a proportion of all residents in households. The score, standardized to all postcode areas in Greater Glasgow, has been linked to mortality, and ranged from -6.25 to 12.87, with higher values indicating higher deprivation.

3.2 Prior Specification

In the Bayesian implementation of the models, we adopted fairly flat normal $N(0.0, 0.001)$ priors for the intercept and fixed effect parameters. However, we assumed prior ignorance on the relative strengths between the unstructured and spatial variance components, and therefore derived their priors based on the observed variability of the log-SMR (Mollie, 1996). These had values of 0.360 and 0.285 for cancer and circulatory disease respectively. Since the variance of the overall relative risk for mortality from cause j in area i is $\sigma_{uj}^2 + \sigma_{ej}^2/n_i$, then, assuming equal relative strengths between the two variances, plausible values for σ_{u1}^2 , σ_{u2}^2 , σ_{e1}^2 and σ_{e2}^2 were 0.180, 0.143, 1.080, and 0.858, respectively, based on 6, the modal number of neighbours for each area. These values were scaled-up by a factor of $\nu = 4$ and entered diagonally on the scale matrix Ω . We know very little about the strength and direction of correlation between the four random effects, thus the off-diagonal elements Ω were set to zero. The specification $\nu = 4$ was used to provide a non-informative prior as it allows maximum variability in the random precision matrix distribution.

3.3 Fitting the models

We fitted two models to the example data: Model A with only the four random effect terms, u_{i1} , u_{i2} , e_{i1} and e_{i2} and model B as Model A including deprivation as an explanatory variable to examine the effect it has on the variance and covariance estimates between the diseases. The frequentist version of the models was estimated using macros written within the *MLwiN* software package, employing quasi-likelihood methods and fitted using iterative generalised least squares procedure (Leyland et al., 2000). In order to speed-up convergence in both methods, the logarithms of the expected counts were centred at 0, thus the intercepts are not equal to 0. For the Bayesian estimation, each model was run using three parallel Gibbs sampler chains for 20,000 iterations from independent starting positions. Convergence of the chains was determined by monitoring all fixed effect and variance parameters using plots of the sample traces, and formally by the Gelman-Rubin criteria. Rapid mixing and convergence was observed by the 1000th iteration. Posterior summaries were based on a combined sample of the last 10,000 iterations, giving a posterior sample size of 30,000.

3.4 Comparison of Results

The results are presented in Table 1. Even though IGLS and WinBUGS are fitting the same models, the former produces the modes of the model parameters based on a quasi-likelihood and from the latter, we report the posterior means (with standard deviations) of the parameters. The fixed effects are shown on the log-relative risk scale and generally are estimated similarly by the two methods. However, the IGLS variance estimates are slightly attenuated compared to those obtained from the Bayesian method. In order to test for the importance of the random effect variances, an upper one-sided z critical

value is used (Self & Liang, 1987). That is, at the 0.05 significance level, the critical value is 1.645. Under Model A, the unstructured heterogeneity and spatial variance z -scores are $0.0113/0.0085 < 1.645$, $0.0084/0.0064 < 1.645$ and $0.0386/0.0289 < 1.645$, $0.0961/0.0350 > 1.645$, respectively, for neoplasms and circulatory disease, under IGLS; and $0.0373/0.0085 > 1.645$, $0.0286/0.0063 > 1.645$ and $0.2393/0.0576 > 1.645$, $0.2040/0.0478 > 1.645$ under the MCMC. This shows that under the IGLS only the spatial circulatory variance is significant whereas using the Gibbs sampler, all of the four variance components are significant. The spatial covariance σ_{e1e2} is significant using IGLS but not under the Gibbs sampler. This is a substantial difference between the two methods, where the IGLS points to evidence of a shared lurking variable and the other says there is none. Furthermore, the relative contributions between the unstructured heterogeneity and spatially structured variance components to the total relative risk variation can also be quantified. In our example, the conditional (on neighbouring areas) contributions of the spatial to unstructured variances are $3.42/n_i$ and $11.44/n_i$ times, respectively, for neoplasms and circulatory disease under IGLS; and $6.42/n_i$ and $7.13/n_i$ times, under the Gibbs sampler. If we replace n_i by 6, then under IGLS, only the circulatory spatially structured component dominates, with a ratio of 1.9 between the conditional spatial and unstructured variances. Under the Gibbs sampler, the ratios of the conditional spatial to unstructured variances are 1.07 and 1.19, showing a slight dominance of the spatial component.

So far, estimates have been based on plugging n_i into the relevant formula, which gives conditional spatial variances and contributions. However, a practical advantage of using the Gibbs sampling method over the IGLS is the ability to generate samples of posterior distribution of parameters. Thus, the behaviour of parameters and functions can be studied over their range

Table 1. Comparison of parameter estimates for the Glasgow mortality data

	IGLS method				Gibbs sampling algorithm			
	Model A		Model B		Model A		Model B	
	Mode	SD	Mode	SD	Mean	SD	Mean	SD
<i>Fixed effects parameters</i>								
Neoplasms								
<i>Intercept</i>	2.8243	0.0326	2.7783	0.0256	2.7960	0.0527	2.7570	0.0572
<i>Deprivation</i>			0.0341	0.0059			0.0328	0.0102
<i>Deprivation²</i>			-0.0016	0.0008			-0.0017	0.0013
Circulatory								
<i>Intercept</i>	3.3903	0.0361	3.3396	0.0318	3.3820	0.0481	3.3380	0.0489
<i>Deprivation</i>			0.0249	0.0060			0.0255	0.0087
<i>Deprivation²</i>			-0.0006	0.0008			-0.0008	0.0011
<i>Random effect parameters</i>								
Neoplasms								
σ^2_{y1}	0.0113	0.0085	0.0018	0.0062	0.0373	0.0085	0.0335	0.0073
σ^2_{e1}	0.0386	0.0289	0.0026	0.0089	0.2393	0.0576	0.2029	0.0248
σ_{u1e1}	0.0209	0.0156	0.0021	0.0074	0.0212	0.0580	0.0091	0.0135
Circulatory								
σ^2_{y2}	0.0084	0.0064	0.0065	0.0058	0.0286	0.0063	0.0268	0.0057
σ^2_{e2}	0.0961	0.0350	0.0495	0.0289	0.2040	0.0478	0.1720	0.0398
σ_{u2e2}	0.0145	0.0110	0.0043	0.0038	0.0140	0.0125	0.0068	0.0107
Cross terms								
σ_{u1u2}	0.0035	0.0048	0.0001	0.0038	0.0020	0.0052	-0.0011	0.0046
σ_{u1e2}	0.0312	0.0121	0.0076	0.0079	0.0137	0.0141	0.0021	0.0123
σ_{e1u2}	0.0065	0.0089	0.0001	0.0045	0.0102	0.0135	0.0015	0.0116
σ_{e1e2}	0.0575	0.0223	0.0091	0.0095	0.0358	0.0365	-0.0016	0.0314

rather than only at the mode. This enables computation of various empirical marginal distributions of, say, spatial variances and their contributions to the total variance. For instance, to assess the relative strength of the spatial and unstructured variations in the data, empirical marginal spatial variances $s_j^2 = \sum_{i=1}^{143} (v_{ij} - \bar{v}_{ij})^2 / (143 - 1)$ were computed at each iteration of the Gibbs sampler chain. The marginal contributions of the spatial effects to the total variability of the random effects were then computed from the resulting sample. The ratios of the marginal spatial to unstructured variances were 1.04 and 1.32 respectively for neoplasms and circulatory disease, indicating that spatially structured component dominates slightly for both causes. These values are slightly similar to the conditional ratios, obtained above, by plugging in the modal number of neighbouring areas.

In Model B, the linear part of the deprivation effect is significant for the two causes under both methods, implying that deprivation is directly linked to mortality from the two causes. Furthermore, the inclusion of deprivation effects has resulted in a general decrease in the estimate of variance parameters. The conditional contributions to the total variability of the random effects by the spatial component have decreased to $1.44/n_i$, $7.64/n_i$ times the heterogeneity part, respectively, for neoplasms and circulatory disease under IGLS; and $6.06/n_i$, $6.45/n_i$ times, respectively, under the Gibbs sampler. The ratios of the marginal spatial to unstructured variances had also decreased to 0.86 and 0.98. This indicates that some of the residual variation shown in Model A is partially accounted for by deprivation, and that greater decreases in the relative importance of the spatial component were observed under the IGLS. The fact that most of the variance-covariance terms are still significant in Model B under the Gibbs sampler, indicates that other unmeasured variables explain the association and variation in the two causes.

Table 2. Results using a different *Wishart* prior

	Model A		Model B	
	Mean	SD	Mean	SD
<i>Fixed effects parameters</i>				
Neoplasms				
<i>Intercept</i>	2.8110	0.0389	2.769	0.0385
<i>Deprivation</i>			0.342	0.0077
<i>Deprivation</i> ²			-0.0017	0.0010
Circulatory				
<i>Intercept</i>	3.3860	0.0386	3.3390	0.0366
<i>Deprivation</i>			0.0259	0.0068
<i>Deprivation</i> ²			-0.0008	0.0008
<i>Random effect parameters</i>				
Neoplasms				
σ_{y1}^2	0.0170	0.0058	0.0119	0.0040
σ_{e1}^2	0.1033	0.0354	0.0655	0.0226
σ_{u1e1}	0.0169	0.0101	0.0028	0.0065
Circulatory				
σ_{y2}^2	0.0150	0.0048	0.0128	0.0040
σ_{e2}^2	0.1202	0.0354	0.0779	0.0248
σ_{u2e2}	0.0130	0.0092	0.0033	0.0070
Cross terms				
σ_{u1u2}	0.0029	0.0037	-0.0007	0.0029
σ_{u1e2}	0.0172	0.0112	0.0021	0.0074
σ_{e1u2}	0.0109	0.0099	0.0021	0.0068
σ_{e1e2}	0.0443	0.0252	0.0025	0.0166

3.5 Sensitivity Analysis

For the Bayesian implementation, our experience has shown that posterior estimates of the covariance matrix Σ can sometimes be overly sensitive to prior specification for the scale Ω . Thus, we performed a sensitivity analysis assuming that the prior random effect variances were smaller than the observed variances of the log-SMRs, which would have over-estimated the order of magnitude of the variance components. In particular, we now assumed that the total variability of the random effect was 0.05 for each cause. Making the

same assumptions as before, this translates into values of 0.025, 0.025, 0.150, and 0.150 for σ_{u1}^2 , σ_{u2}^2 , σ_{e1}^2 and σ_{e2}^2 , which were scaled-up by a factor of $\nu = 4$ and then entered diagonally into the scale matrix Ω , with off-diagonal elements still set to 0. The results of this sensitivity analysis are presented in Table 2. While posterior summaries for the regression (deprivation) effects are relatively unaffected, inferences for the variance components are slightly attenuated. Despite this, conclusions concerning both the regression and variance components were not affected, thus we are satisfied with our original prior specification of the hierarchical precision prior Σ^{-1} .

4. Discussion

Two different estimation methods: iterative generalised least squares (IGLS) estimation and Gibbs sampler, a sampling-based Bayesian approach have been used to simultaneously estimate relative risk of mortality from two-disease cause. These two methods offer important alternative estimation procedures to many estimation methods such as the EM algorithm. The ability to model and estimate multiple disease risks offers many advantages over the usual univariate estimation, in particular inferences can be made on the covariance between the diseases before and after allowing for possible spatial covariates. Parameter estimates of the fixed covariate effect are similar between the two methods, but differ on the need to include random effects. Moreover, it has been shown that IGLS variance parameter estimates are attenuated compared to the Gibbs sampler and that the latter produces variance estimates, which are markedly affected by the prior specifications for the random effect precision matrix. These results are consistent with previous comparisons between a frequentist and a full Bayesian approach to fitting univariate spatial disease models.

The IGLS procedure is essentially an empirical Bayesian estimation as the parameters describing the random effects are estimated from the observed data. For instance, the random effects themselves are estimated as posterior means conditioned on the data and “plug-in values” (usually maximum likelihood estimates) of the hyperparameters describing the random effect distribution. Thus, the variability of the random effect estimates are underestimated because the uncertainty in the hyperparameters is not accounted for (Mollie, 1996). In a full Bayesian estimation all the uncertainty in the parameters is modelled. The inclusion of prior distribution of model parameters has caused great controversy on the grounds that too much subjective prior information may unnecessarily influence the results. However, Dunson (2001) argues that by allowing prior knowledge in the analysis, full Bayesian methods offer a broader class of models, whose robustness and sensitiveness can be checked by adopting different priors for different parts of the model. In using Gibbs sampling, samples of posterior distributions of parameters can be obtained unlike in the IGLS procedure. Thus, the posterior behaviour of parameters and their functions could be studied over their range rather than only at the mode. For instance, to assess the relative strength of unstructured and spatially structured variance components in the model, requires estimates of the empirical marginal, rather than conditional variances. The marginal contribution is easily computed at each run of a Bayesian MCMC, which provides a sample of the contributions. The resultant sample can provide more informative summaries than point estimates obtained from plugging in modes or median values into relevant formulae. However, in a recent review of empirical Bayesian estimation in univariate disease mapping, Leyland and Davies (2005) argue for both empirical and full Bayesian methods.

The research so far has specified the interdependence between disease outcomes through a multivariate normal model for regression errors, which are not directly observed. This is largely for convenience purposes as many statistical packages nowadays routinely handle inferences based on the multivariate normal distribution. Thus, the use of independent Poisson models, with a sensible MCAR prior on the random effects seems to be the direction to handle data of this sort. However, the natural modelling choice for truly multivariate count is the multivariate Poisson distribution as in Karlis (2003); however, the multivariate Poisson distribution has rarely been used in spatial epidemiology. Multivariate Poisson models may also help solve current difficulties in specifying the correlation structure between each pair of diseases in the same and different areas.

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References

- ASSUNÇÃO, R.M. & CASTRO, M.S.M. (2004). Multiple cancer sites incidence rates estimation using multivariate Bayesian model. *International Journal of Epidemiology* **33**, 508–516.

- BESAG J, YORK, J. & MOLLIE, A. (1991). Bayesian image restoration, with two applications in spatial statistics (with discussion). *Annals of the Institute of Statistical Mathematics* **43**, 1–75.
- BROWNE, W.J., GOLDSTEIN, H. & RASBASH, J. (2001). Multiple membership multiple classification (MMMC). *Statistical Modelling*, **1**, 103–124.
- CLAYTON, D. & KALDOR, J. (1987). Empirical Bayes estimates of age-standardized relative risks for use in disease mapping. *Biometrics* **43**, 671–81.
- DABNEY, A.R. & WAKEFIELD, J.C. (2005). Issues in the mapping of two diseases. *Statistical Methods in Medical Research*, **14**, 83–112.
- DUNSON, D.B. (2001). Commentary: Practical advantages of Bayesian analysis of epidemiologic data. *American Journal of Epidemiology*, **153**, 1222–1226.
- FENN, B., MORRIS, S.S. & BLACK, R.E. (2005). Comorbidity in childhood in northern Ghana: Magnitude, associated factors, and impact on mortality. *International Journal of Epidemiology*, **34**, 368–375.
- FELTBOWER, R.G., MANDA, S.O.M., GILTHORPE, M.S., GREAVES, M.F., PARSLow, R.C., KINSEY, S.E., BODANSKY, H.J. & McKINNEY, P.A. (2005). Detecting small area similarities in the epidemiology of childhood acute lymphoblastic leukemia and type 1 diabetes: a Bayesian approach. *American Journal of Epidemiology*, **161**, 1168–1180.
- HEISTERKAMP, S.H., DOORNBOS, G. & GANKEMA, M. (1993). Disease mapping using empirical Bayes and Bayes methods on mortality statistics in The Netherlands. *Statistics in Medicine*, **12**, 1895–1913.
- HELD, L., NATARIO, I., FENTON, S.E., RUE, H. & BECKER, N. (2005). Towards joint disease mapping. *Statistical Methods in Medical Research*, **14**, 61–82.
- JIN, X., CARLIN, B.P. & BANERJEE, S. (2005). Generalised hierarchical multivariate CAR models for areal data.
- KARLIS, D. (2002). An EM algorithm for multivariate Poisson distribution and related models. *Journal of Applied Statistics*, **30**, 63–77. *Biometrics*, **6**, 539–557.

- KIM, H., SUN, D. & TSUTAKAWA, R.K. (2001). A bivariate method for improving the estimates of mortality rates with a twofold conditional autoregressive model. *Journal of the American Statistical Association*, **96**, 1506–21.
- KNORR-HELD, L. & BEST, N.G. (2001). A shared model for detecting joint and selective clustering of two diseases. *Journal of the Royal Statistical Society, Series A*, **164**, 73–86.
- LANGFORD, I.H., LEYLAND, A.H., RASBASH, J. & GOLDSTEIN, H. (1999). Multilevel modelling of geographical distributions of rare diseases. *Journal of the Royal Statistical Society, Series C*, **48**, 253–269.
- LEYLAND, A.H., LANGFORD, I.H., RASBASH, J. & GOLDSTEIN, H. (2000). Multivariate spatial models for event data. *Statistics in Medicine*, **19**, 2469–2478.
- LEYLAND, A.H. & DAVIES, C.A. (2005). Empirical Bayes methods for disease mapping. *Statistical Methods in Medical Research*, **14**, 17–34.
- LIU, X., WALL, M.M. & HODGES, J.S. (2005). Generalised spatial structural equation models. *Biostatistics*. To appear.
- MACNAB, Y.C., FARRELL, P.J., GUSTAFSON, P. & WEN, S. (2004). Estimation in Bayesian disease mapping. *Biometrics*, **60**, 865–873.
- MANDA, S.O.M. (2006). Bayesian bivariate diseases mapping models using two-cause mortality data from Greater Glasgow, Scotland. *Biometrics*, **60**, 865–873.
- MARDIA, K.V. (1988). Multi-dimensional multivariate Gaussian Markov random fields with application to image processing. *Journal of Multivariate Analysis*, **24**, 265–284.
- MOLLIE, A. (1996). Bayesian mapping of disease. In *Markov Chain Monte Carlo in Practice* (ed. W.R. Gilks, S. Richardson & D.J. Spiegelhalter), pp. 359–379. Chapman and Hall.
- RASBASH, J., STEELE, F., BROWNE, W. & PROSSER, B. (2004). *A User's Guide to MLwiN Version 2.0*. Centre for Multilevel Modelling, Institute of Education: London (2004).

- SELF, S.G. & LIANG, K. (1987) Asymptotic properties of maximum likelihood estimators and likelihood ratio test under nonstandard conditions. *Journal of the American Statistical Association*, **82**, 605–610.
- SMITH, A.F.M. & ROBERTS, G.O. (1993). Bayesian computations via the Gibbs sampler and related Markov chain Monte Carlo methods (with discussion). *Journal of the Royal Statistical Society, Series B*, **55**, 1993; 3–24.
- SPIEGELHALTER, D.J., THOMAS, A., BEST, N.G. & LUNN, D. (2004). *BUGS: Bayesian Inference Using Gibbs Sampling, Version 1.4.1*. Medical Research Council Biostatistics Unit, Cambridge University.
- WANG, F. & WALL, M.M. (2003). Generalized common spatial factor model. *Biostatistics*, **4**, 569–582.